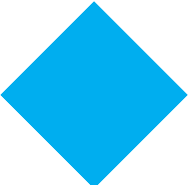




Smart Robotics Project



... Air Hockey ...

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Project Idea

Our reference: Robot Air Hockey Challenge

Robot
Franka Emika
Panda



- ◆ Software: ROS & Python 3.8
- ◆ Camera-based vision system
- ◆ Real-time puck tracking
- ◆ Future puck position prediction

System Setup



GAZEBO | Physics-based robot simulation

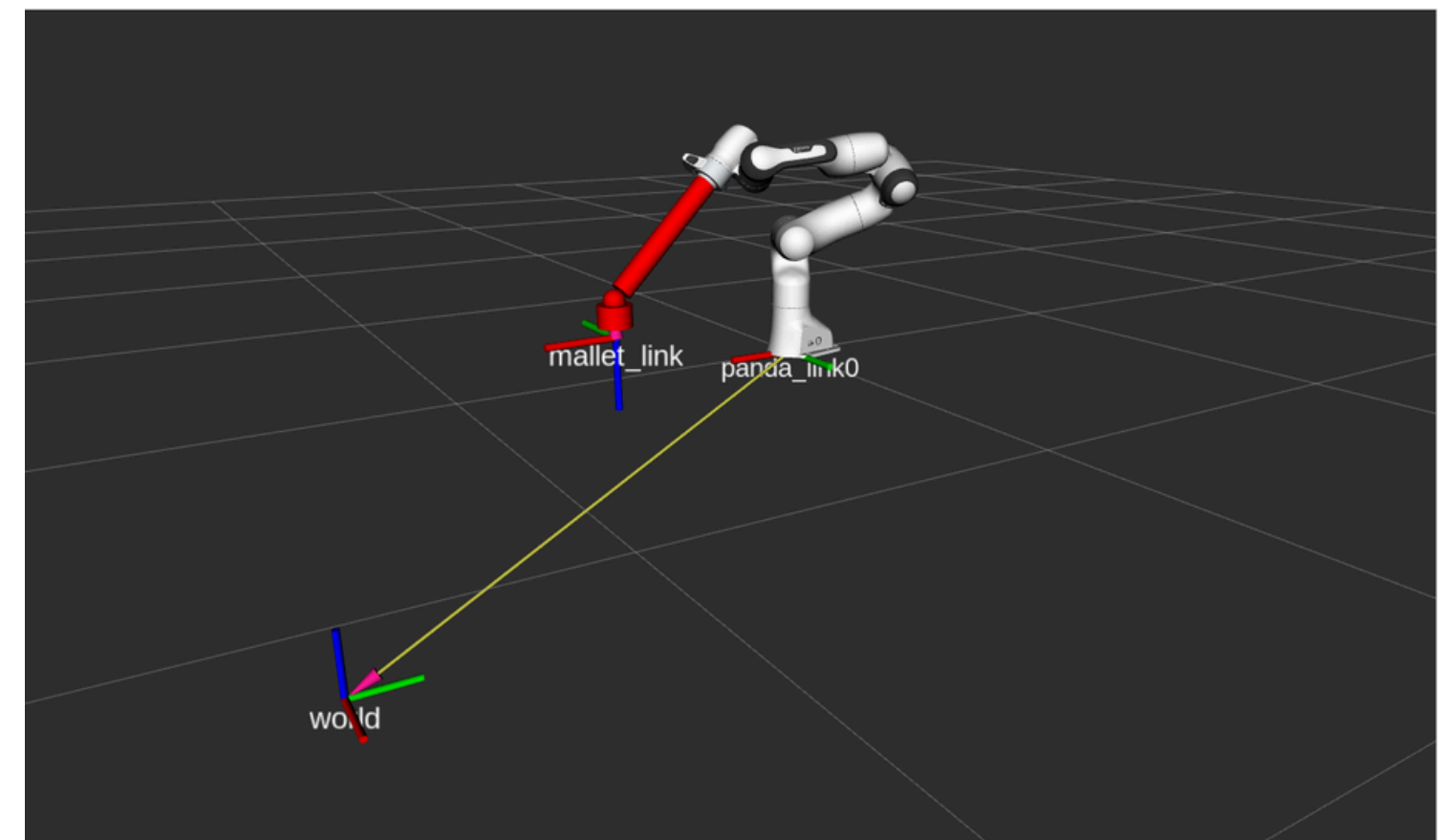
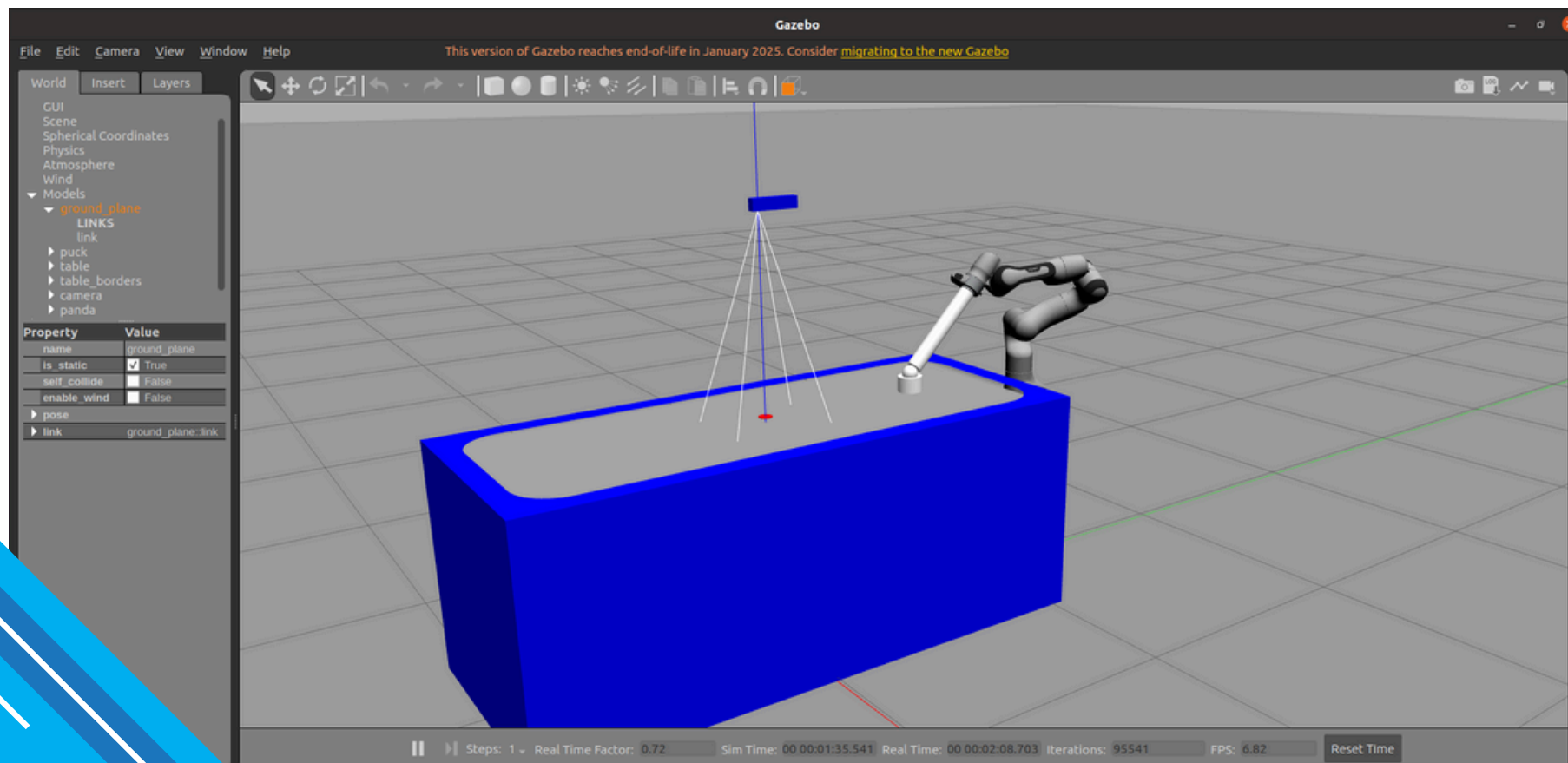
Hardware safety guaranteed

Usefull for prototyping

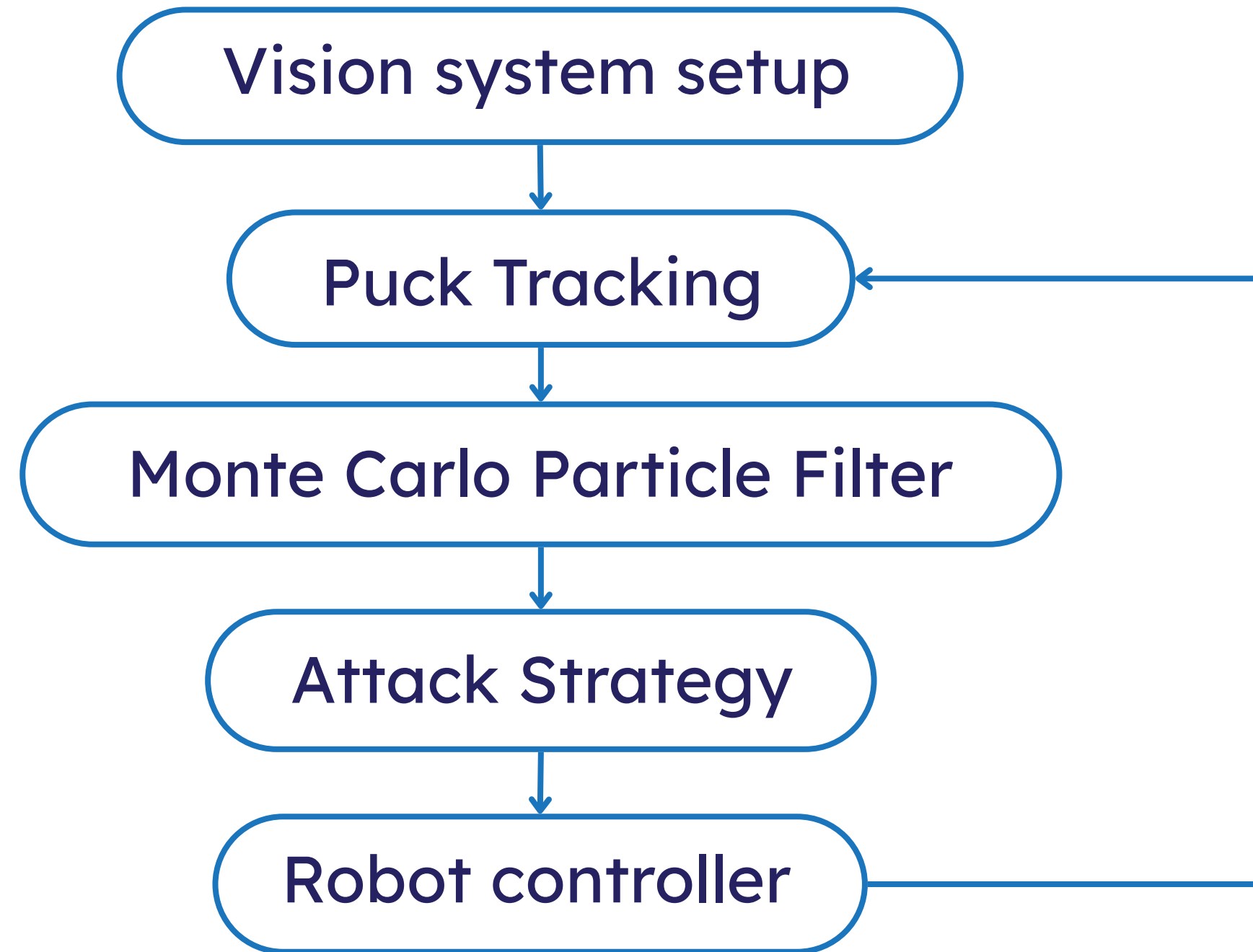
RVIZ | ROS visualization tool

Visualizes robot states and sensor data

Can issue movement commands



Pipeline





Vision system setup

Our OpenCV-based pipeline

Barrel Distortion correction

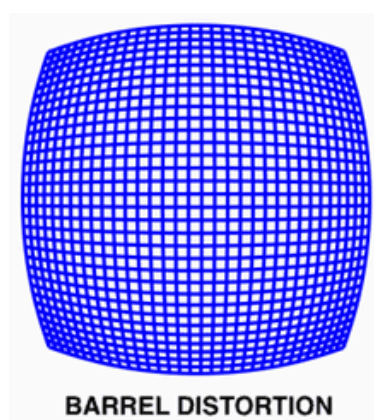


Table corner detection pipeline

HSV white masking
↓
Canny edge detection
↓
Hough transform

Homography computation

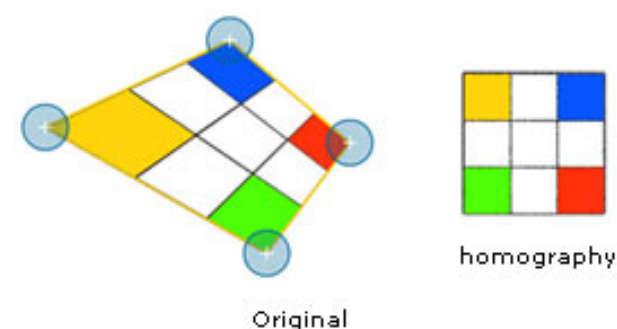
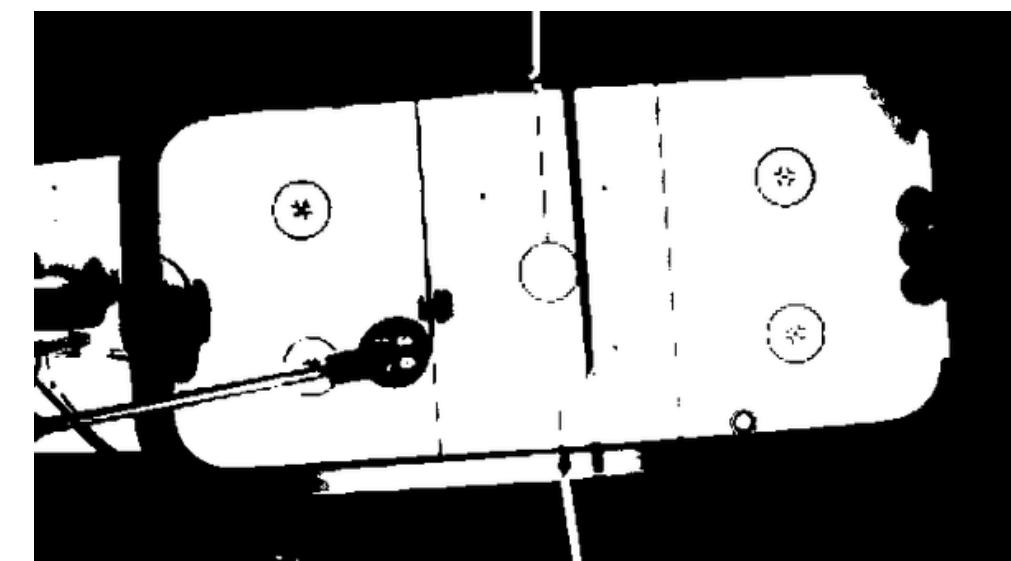


Table without barrel distortion



HSV white masking



Table corner markers

1

Puck Tracking



Puck detection

Our OpenCV-based
pipeline



Puck Detection

HSV red masking



Circle detection using
connected components



Puck center extraction

pixel-meter
Conversion

Homography-based coordinate conversion

Image-to-world mapping

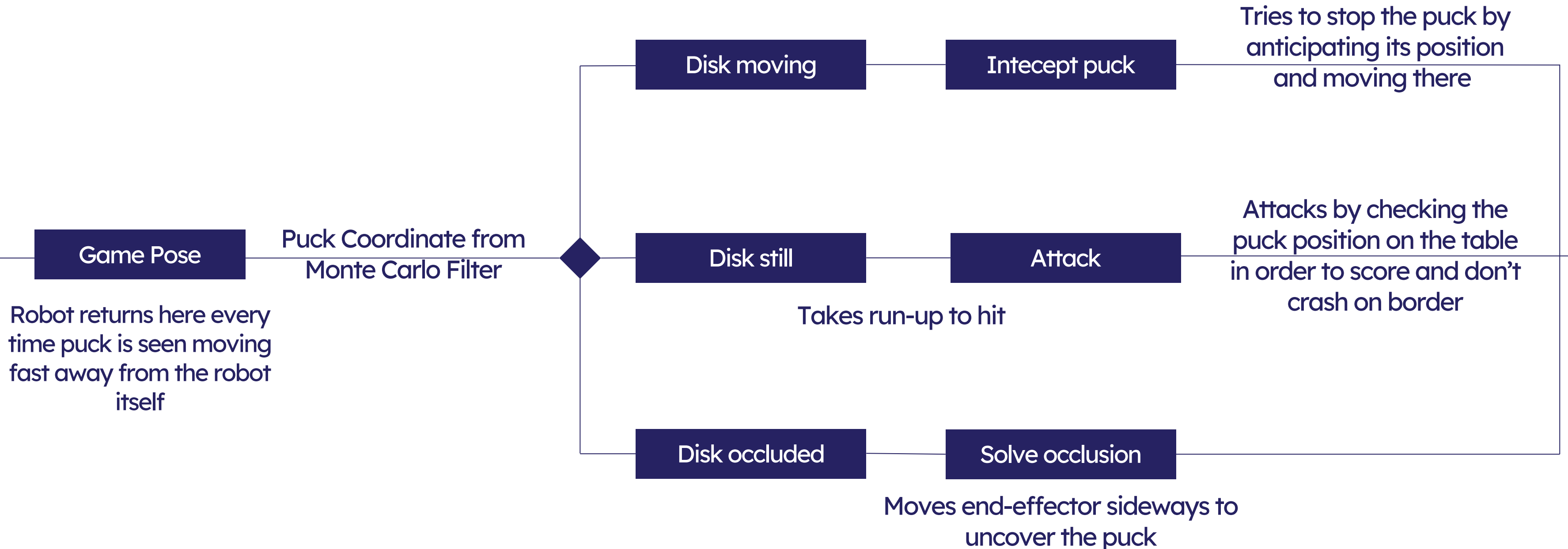


Monte Carlo Particle Filter





Attack Strategy



4

Robot controller



Our initial control:

Motion Planning Framework

Limitation: high latency

Recommended:

Cartesian Pose Controller

ROS-based motion planning framework

Trajectory planning and execution

Integrated with RViz for interaction

Real-time end-effector control in Cartesian space

Commands position and orientation directly (x, y, z, R)

Operates at high control frequency

franka-ros

ROS integration of Franka robots
Provides controllers and ROS interfaces

Franka Control Interface (FCI)

Communication layer between PC and robot
Ensures real-time data exchange

libfranka

Low-level C++ control interface
Direct access to robot hardware



Future Improvements

- ◆ Integrate an impedance controller
to enable compliant and safe interaction with the environment and humans
- ◆ Explore advanced prediction methods
Reinforcement Learning (RL)
LSTM networks
- ◆ Migrate from ROS 1 to ROS 2
improved real-time performance and reliability

Smart Robotics Project

... **Thank you** ...

[Link GitHub](#)

